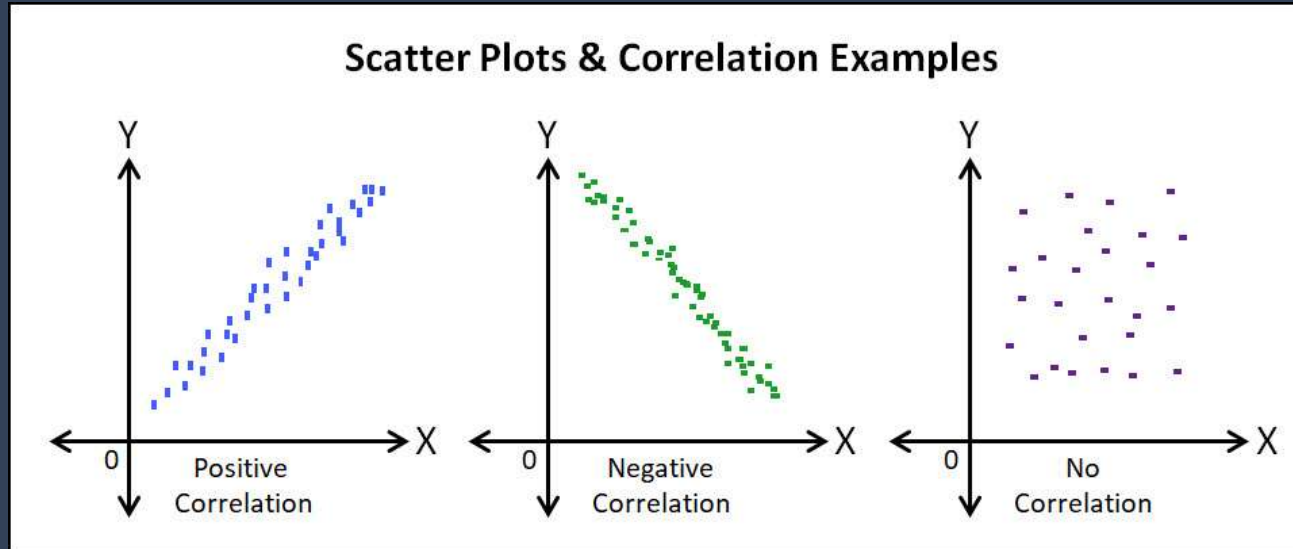


Correlation

Correlation

- As mentioned, covariance can give you information about the direction of the relationship between two variables but not about the strength of the relationship.
- On the other hand, correlation indicated the strength of the relationship between two variables
- The correlation coefficient can take values between -1 and +1.
- A negative correlation coefficient implies that one variable increases as the other decreases, while a positive coefficient implies that both variables increase or decrease together (0 indicates no relationship)

Correlation



Correlation

Correlation coefficient:

$$r_{XY} = \sum (x_i - \mu_x)(y_i - \mu_y) / (\sum (x_i - \mu_x)^2 (y_i - \mu_y)^2)^{1/2}$$

This equation is normalized and constrained between -1 and +1, and hence is indicative of the strength of the relationship between two variables

Do correlation and dependence mean the same thing?

Correlation

Do correlation and dependence mean the same thing?

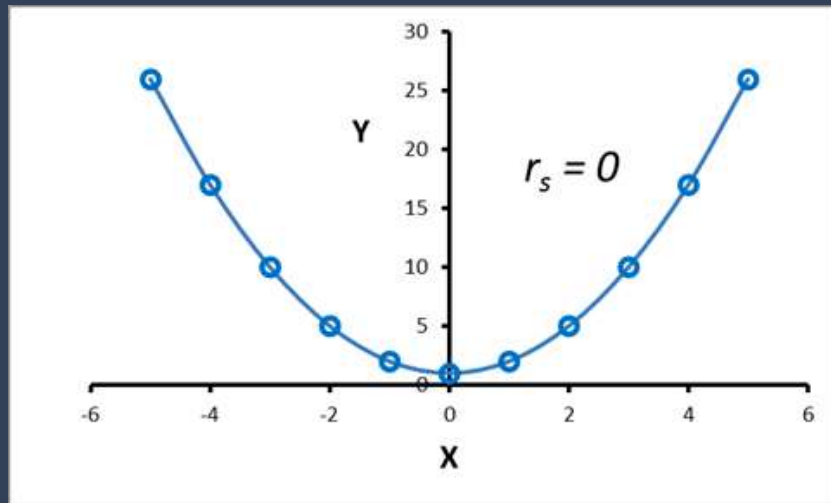
A non-dependence between two variables $\Rightarrow$ 0 correlation

However, a zero correlation does not necessarily mean non-dependence.

Correlation can only capture linear relationship between variables. However, if two variables are related in a non-linear fashion, they could have a zero correlation. Hence, zero correlation does not necessarily mean non-dependence between variables

Correlation

Example: Dependent variables with 0 correlation



Perfect quadratic relationship

Correlation

Q2: Calculate covariance and correlation for the following data set:

X: 3, 6, 10, 12 (= 7.75)

Y: 8, 10, 12, 14 (= 11)

Covariance:

$\text{Cov}(X, Y)$

$$\begin{aligned} &= (3-7.75)(8-11) + (6-7.75)(10-11) + (10-7.75)(12-11) + (12-7.75)(14-11) / (4-1) \\ &= 10.33 \end{aligned}$$

Correlation

Q2: Calculate covariance and correlation for the following data set:

X: 3, 6, 10, 12 (= 7.75)

Y: 8, 10, 12, 14 (= 11)

Correlation:

$$\text{Cov}(X, Y) = r_{XY} = \sum (x_i - \mu_x)(y_i - \mu_y) / (\sum (x_i - \mu_x)^2 (y_i - \mu_y)^2)^{1/2}$$

Numerator: $(3-7.75)(8-11)+(6-7.75)(10-11)+(10-7.75)(12-11)+(12-7.75)(14-11) = 30.99$

Denominator: $\text{sqrt}((3-7.75)^2(8-11)^2+(6-7.75)^2(10-11)^2+(10-7.75)^2(12-11)^2+(12-7.75)^2(14-11)^2) = 715.40$

Hence, **Corr(X, Y) = 0.04**